Question 6 Below the Critical Exponent: Cantor-Product Embeddings Force Undecidable Semilocalizability

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9 August 2026

Status and scope

Candidate strong partial resolution, likely valid, pending specialist review. Question 6 of De Pauw asks, for $0 < d < 1$ and $d \neq \frac{1}{2}$, whether

$$([0,1], \mathcal{A}_{\mathcal{H}^d}, \mathcal{H}^d)$$

is consistently not semilocalizable [1]. This packet answers the question affirmatively for every $0 < d < \frac{1}{2}$. Combined with the source paper's theorem at $d = \frac{1}{2}$, this settles the whole lower range $0 < d \leq \frac{1}{2}$. No claim is made for $\frac{1}{2} < d < 1$.

The new step is to place a bi-Lipschitz copy of the product $C_d \times C_d$ inside $[0,1]$ when $d < \frac{1}{2}$. De Pauw's vertical–horizontal theorem then gives consistent nonlocalizability on this compact copy. A trace and principal-ideal argument transfers the obstruction to the whole interval.

Source question

The question occurs in Section 11.6, printed and PDF page 30 of arXiv:1909.10190v2. The crop includes the explanation of the product structure and the complete statement of Q6.

This boils down to deciding whether the σ-algebra $\mathscr{A}$ defined in the proof of 9.8 coincides with the σ-algebra $\mathscr{A}_{\mathscr{H}^d}$. The fact the answer to this question is not known turned out to be no obstacle thanks to the freedom allowed in 8.3 regarding the σ-algebra $\mathscr{A}$.

11.6. — Our last question here concerns 9.8. The proof, based on 8.3 seems to require a product structure that forces the dimension to be $1/2$.

(Q6) *Let $0 < d < 1$ and $d \neq \frac{1}{2}$. Is the measure space $\left([0,1], \mathscr{A}_{\mathscr{H}^d}, \mathscr{H}^d\right)$ consistently not semilocalizable?*

Transcription: for $0 < d < 1$, $d \neq \frac{1}{2}$, is the measure space $([0,1], \mathcal{A}_{\mathcal{H}^d}, \mathcal{H}^d)$ consistently not semilocalizable?

Definitions and source inputs

For an outer measure μ on X, let $\mathcal{A}_\mu$ be its Caratheodory σ-algebra and put

$$\mathcal{A}_\mu^f = \{F \in \mathcal{A}_\mu : \mu(F) < \infty\}, \qquad \mathcal{N}_\mu^{\mathrm{loc}} = \{N \in \mathcal{A}_\mu : \mu(N \cap F) = 0 \text{ for every } F \in \mathcal{A}_\mu^f\}.$$

A measurable space with negligibles $(X, \mathcal{A}, \mathcal{N})$ is *localizable* when every family in $\mathcal{A}$ has an essential supremum modulo $\mathcal{N}$; equivalently, the quotient Boolean algebra $\mathcal{A}/\mathcal{N}$ is order complete. A measure space $(X, \mathcal{A}, \mu)$ is *semilocalizable* when $(X, \mathcal{A}, \mathcal{N}_\mu^{\mathrm{loc}})$ is localizable [1].

We use two results proved in the source paper.

(D1) Under the relatively consistent inequality

$$\mathrm{non}(\mathcal{N}_{\mathcal{L}^1}) < \mathrm{cov}(\mathcal{N}_{\mathcal{L}^1}),$$

De Pauw's Theorem 8.3 says that $(X, \mathcal{A}, \mathcal{N})$ is not localizable if it admits two Polish diffuse probability parameter spaces S, T and measurable sets V_s, H_t such that:

- (a) $\varnothing \neq V_s \cap H_t \in \mathcal{N}$ for every s, t;
- (b) if $V_s \cap Z \in \mathcal{N}$, then the outer probability of $\{t : H_t \cap V_s \cap Z \neq \varnothing\}$ is zero;
- (c) the symmetric assertion holds after exchanging V, S with H, T.

(D2) Under the Continuum Hypothesis, every Borel-regular outer measure on a Polish space has the almost-decomposition property used in the paper; this implies semilocalizability (Propositions 5.3–5.4 of the source).

Proof intuition

Let C_d be a two-branch Cantor set whose contraction ratio is $r = 2^{-1/d}$. The identity $2r^d = 1$ gives its Hausdorff dimension d. When $d < \frac{1}{2}$, one has $4r < 1$, so four separated copies of ratio r fit inside an interval. Pair the two binary digits of a point of $C_d \times C_d$ into one four-valued digit. This produces a bi-Lipschitz coding of the product onto a compact four-branch Cantor set $K_d \subset [0, 1]$.

The images of the vertical and horizontal leaves are d-dimensional sets of finite positive measure and intersect in exactly one point. If a measurable set is locally null on one leaf, it is genuinely $\mathcal{H}^d$-null there; the opposite parameter incidence set is a Lipschitz image and is therefore null. This is precisely the input required by (D1).

Finally, the measurable quotient for K_d is the principal ideal below the class of K_d in the interval quotient. Principal ideals of complete Boolean algebras are complete, so localizability of the interval would force localizability of K_d, a contradiction.

The Cantor-product embedding

Fix $0 < d < \frac{1}{2}$ and set

$$r = 2^{-1/d}, \qquad g_C = 1 - 2r, \qquad g_K = \frac{1 - 4r}{3}.$$

Thus $0 < r < \frac{1}{4}$ and $g_C, g_K > 0$. Let C_d be the attractor of

$$S_0(x) = rx, \qquad S_1(x) = 1 - r + rx,$$

and let K_d be the attractor of the four similarities

$$P_a(x) = rx + \frac{a(1-r)}{3}, \qquad a = 0, 1, 2, 3.$$

The four first-generation intervals are disjoint, with adjacent gap g_K.

Lemma 1 (Digit pairing is bi-Lipschitz)**.** *There is a bi-Lipschitz bijection*

$$\Phi : C_d \times C_d \longrightarrow K_d.$$

With the Euclidean product metric it satisfies

$$\frac{g_K}{\sqrt{2}}\,\|(x,y)-(x',y')\|_2 \le |\Phi(x,y)-\Phi(x',y')| \le \frac{1}{g_C}\,\|(x,y)-(x',y')\|_2.$$

Proof. Strong separation gives every point of C_d a unique binary address. If x and y have addresses (i_n) and (j_n), define $\Phi(x,y)$ to be the point of K_d with address $(2i_n+j_n)$. This is a bijection because digit pairing is a bijection from $\{0,1\}^2$ to $\{0,1,2,3\}$ at every level.

Suppose two paired addresses first differ at level n. On C_d, two binary addresses first differing at level n have distance between $g_C r^{n-1}$ and r^{n-1}. Consequently, on $C_d \times C_d$,

$$g_C r^{n-1} \le \|(x,y)-(x',y')\|_2 \le \sqrt{2}\, r^{n-1}.$$

On K_d, two four-valued addresses first differing at level n have distance between $g_K r^{n-1}$ and r^{n-1}. Combining these two pairs of bounds gives the displayed inequalities. □

The open set condition gives

$$0 < \mathcal{H}^d(C_d) < \infty, \tag{1}$$

since $2r^d = 1$. Notice also that $\dim_H K_d = 2d$; the point of the construction is not that K_d itself has finite $\mathcal{H}^d$ measure, but that its two transverse leaf families do.

The vertical–horizontal obstruction on K_d

For $s,t \in C_d$, define compact subsets of K_d by

$$V_s = \Phi(\{s\}\times C_d), \qquad H_t = \Phi(C_d \times \{t\}).$$

Lemma 2 (The source theorem applies on K_d)**.** *Let $\phi = \mathcal{H}^d|_{K_d}$, $\mathcal{A} = \mathcal{A}_\phi$, and $\mathcal{N} = \mathcal{N}_\phi^{\mathrm{loc}}$. Under* $\mathrm{non}(\mathcal{N}_{\mathcal{L}^1}) < \mathrm{cov}(\mathcal{N}_{\mathcal{L}^1})$*, the measurable space with negligibles $(K_d, \mathcal{A}, \mathcal{N})$ is not localizable.*

Proof. By Lemma 1 and (1), every V_s and H_t is compact and has finite positive $\mathcal{H}^d$ measure. Moreover

$$V_s \cap H_t = \{\Phi(s,t)\},$$

which is nonempty and $\mathcal{H}^d$-null because $d > 0$.

Give C_d the normalized diffuse Borel probability $\mathcal{H}^d|_{C_d}/\mathcal{H}^d(C_d)$. Fix $Z \in \mathcal{A}$. If $V_s \cap Z \in \mathcal{N}$, then the finite-measure test set V_s in the definition of local nullity yields

$$\mathcal{H}^d(V_s \cap Z) = 0.$$

The map $V_s \to C_d$, $\Phi(s,t) \mapsto t$, is Lipschitz: it is a coordinate projection after the Lipschitz inverse of Φ. Hence

$$\mathcal{H}^d(\{t \in C_d : H_t \cap V_s \cap Z \ne \varnothing\}) = 0.$$

The same argument with the coordinates exchanged handles a locally null $H_t \cap Z$. Thus all hypotheses of De Pauw's Theorem 8.3 are satisfied, and input (D1) proves the claim. □

Restriction to a measurable subspace

Lemma 3 (Trace and principal ideal)**.** *Let $\mu = \mathcal{H}^d$ on $[0,1]$, let $K \subset [0,1]$ be Borel, and let $\phi = \mu|_K$ as an outer measure on the metric subspace K. Then*

$$\mathcal{A}_\phi = \{A \cap K : A \in \mathcal{A}_\mu\},$$
$$\mathcal{N}_\phi^{\mathrm{loc}} = \{N \cap K : N \in \mathcal{N}_\mu^{\mathrm{loc}}\}.$$

Consequently $\mathcal{A}_\phi/\mathcal{N}_\phi^{\mathrm{loc}}$ is isomorphic to the principal ideal below $[K]$ in $\mathcal{A}_\mu/\mathcal{N}_\mu^{\mathrm{loc}}$.

Proof. Hausdorff outer measure on the metric subspace agrees with ambient Hausdorff outer measure on subsets of K. The trace of an ambient measurable set is therefore ϕ-measurable. Conversely, suppose $E \subset K$ is ϕ-measurable. Since the Borel set K is μ-measurable, for arbitrary $S \subset [0,1]$ we have

$$\begin{aligned}\mu(S) &= \mu(S \cap K) + \mu(S \setminus K) \\ &= \mu(S \cap E) + \mu((S \cap K) \setminus E) + \mu(S \setminus K) \\ &= \mu(S \cap E) + \mu(S \setminus E).\end{aligned}$$

Thus $E \in \mathcal{A}_\mu$, proving the first identity.

If $N \subset K$ is locally null in the ambient space, it is locally null in K. Conversely, if it is locally null in K and $F \in \mathcal{A}_\mu$ has finite measure, then $F \cap K$ is a finite-measure member of $\mathcal{A}_\phi$ and

$$\mu(N \cap F) = \phi(N \cap (F \cap K)) = 0.$$

This proves the second identity. The map $[E] \mapsto [E]$ identifies the subspace quotient with the classes below $[K]$: if $[A] \leq [K]$, then $A \setminus K$ is locally null and $[A] = [A \cap K]$. Hence the image is exactly the principal ideal below $[K]$. ☐

Lower-half resolution

Theorem 1. *For every $0 < d < \frac{1}{2}$, the statement*

$$([0,1], \mathcal{A}_{\mathcal{H}^d}, \mathcal{H}^d) \text{ is semilocalizable}$$

is undecidable in ZFC, in the usual relative-consistency sense. More precisely:

(i) *under the Continuum Hypothesis the space is semilocalizable;*

(ii) *under $\mathrm{non}(\mathcal{N}_{\mathcal{L}^1}) < \mathrm{cov}(\mathcal{N}_{\mathcal{L}^1})$ it is not semilocalizable.*

Proof. Part (i) is input (D2), applied to the Borel-regular outer measure $\mathcal{H}^d$ on the Polish space $[0,1]$.

For (ii), let K_d be the compact set constructed above. Lemma 2 says that the quotient associated with K_d is not order complete. If the interval were semilocalizable, then

$$\mathcal{A}_{\mathcal{H}^d}([0,1])/\mathcal{N}_{\mathcal{H}^d}^{\mathrm{loc}}([0,1])$$

would be order complete. Every principal ideal in an order-complete Boolean algebra is order complete: the supremum of a family below b is the ambient supremum, which is still below b. Lemma 3 identifies the quotient for K_d with the principal ideal below $[K_d]$, giving a contradiction. Therefore the interval is not semilocalizable under the stated consistent inequality. ☐

Corollary 1. *Question 6 has an affirmative answer for every $0 < d < \frac{1}{2}$. Together with De Pauw's Theorem 9.8 at $d = \frac{1}{2}$, undecidable semilocalizability is known throughout $0 < d \leq \frac{1}{2}$.*

Why the same route stops above one half

The construction has a sharp geometric barrier. Four separated first-generation intervals of ratio $r = 2^{-1/d}$ fit in an interval only if $4r \le 1$, equivalently $d \le \frac{1}{2}$. More invariantly,

$$\dim_H(C_d \times C_d) = 2d > 1 \qquad (d > \tfrac{1}{2}),$$

so no bi-Lipschitz copy of the product can lie in $\mathbb{R}$. This does not show that Q6 is false in the upper half. It shows only that any upper-half proof must replace the equal-dimensional product-incidence geometry by a different mechanism.

Novelty check and review status

A bounded search of the run indexes, exact-title web/arXiv results, the source paper's publication record, and later work on localizable locally determined measurable spaces found no later paper explicitly answering Q6. The proof is short because its new ingredient is an elementary geometric transport of the source paper's own Theorem 8.3. Novelty remains provisional until specialist literature review.

The main verifier points are the uniform first-different-digit estimates, the passage from local nullity to nullity on finite-measure leaves, and the trace/principal-ideal identification. Each is proved explicitly above.

References

[1] T. De Pauw, *Undecidably semilocalizable metric measure spaces*, Commun. Contemp. Math. **26** (2024), no. 4, 2350012; arXiv:1909.10190, https://arxiv.org/abs/1909.10190.